

Digital transformation of Human Resources Management: A Roadmap

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Abstract—Digital transformation aims to help Human Resources (HR) management to achieve his strategic goals. It improves the time, the costs, the methods and the quality of HR services. The modern company performs digital transformation for practically all HR activities, hiring, compensation, assessment, Workforce planning and training. In this paper, we aim to determine the different digital transformation types in the HR field. Hence, this article is an attempt to identify and analyze seven HR digital transformation roadmaps (Four are specifics to e-HRM, Smart HR 4.0, HRIS and Cloud computing). In our study, we try to know and describe the principal steps and the essential elements of a roadmap in this field.

Keywords— HR management, digital transformation, roadmap, e-HRM, industry 4.0, HRIS.

I. INTRODUCTION

The Covid-19 pandemic context, has accelerated and extended the digital transformation of organizations [1] [2] [3]. The uses of many types of information and communication technologies (ICT) have enabled companies to maintain their activities. The pandemic is forcing the companies to work remotely, and to accelerate digital technology implementation [1].

In a context such as that of Covid-19, the adoption of new digital technology is the right solution to maintain the organisations sustainability [4]. It is one of the most important challenges that organizations currently face. No company or activity is immune to the digital technology effects [5]. It is changing all economic areas [6]. The digital technology adoption has several concepts such as digital transformation, digitalization and digitization [7].

The concept "digital transformation" means the networking based on advanced technological development of all company's actors across the levels of the value chain [6]. It is the adoption of digital technology with the but of improving the organization's performance [4].

HR management is a system of strategies, proceedings and functions that focus on managing human capital and the employment relationship at all levels of a company in order to realize the organisational objective [8]. It is a strategic approach for an efficient employment and progress of a highly engaged and skilled employees. HR management influences the performance of the organization and development. The essential functions of HR management are the employee

selection, recruitment, training [9], compensation [10], and performance appraisal [8].

The digital transformation of HR management signifies the technical process of transforming manual HR information into digital form. It is the operation by utilizing digital potentials for strategic and operational HR management goals [7]. Digital technologies HR is a group of tools (HRIS, electronic pointing, chatbots, gamification, etc.), services and techniques used to optimize HR management and to facilitate company transformations [11].

The main goal of HR digitalization is to achieve HR operations and employees register keeping more accurately and efficiently. It can deliver advice to HR managers in various areas [12], and impact HR practices (Datafication) and HR management (Culture) [11]. The digitalization of HR management is also called as e-HRM. The important types of e-practices in the HR field are e-compensation, e-recruitment, e-training [13].

The technology roadmap provides orientation for aware policy-making. The roadmap is employed for forecasting the future technology and recognizing the best possibilities for a competitive edge [14].

Many authors are focused on researching how digitalization has streamlined HR processes such as recruitment, performance management and HR planning [13] [15] [16] [17] [18]. Several roadmaps have been proposed in HR digital transforming and successfully implementing of the ICT and systems in the HR management field [19] [20] [21] [22] [23].

This paper is structured as follows. First, we present the principal roadmap concepts. Then we identify the main HR digitalization types, and we define the methodology of our study. After that, we analyze some specific and general roadmaps of HR digital transformation, and, finally, we synthesize the main common roadmaps elements.

II. THEORETICAL BACKGROUND

A. Roadmap concept

A roadmap is a good planning and a prospective look of a specific domain. It is based on the collective knowledge and the pertinent change elements [24]. Abbasi et al. [14] define roadmap as a means for strategic planning, and a tool of knowledge foundation for company stakeholders. It is used for integrating organization strategy, technology, Research and development activities, and product development.

Issa et al. [25] mentioned that companies can create their own roadmaps for improving the outcome of the Industry 4.0 implementation and attaining a higher technological maturity level. Camarihna-Matos et al. [26] added that roadmap is a result of the stakeholders imagination and the change drivers knowledge, it gives an extensive look of current and prospective ageing.

B. Human Resource Information Systems (HRIS)

HRIS as “the IT applications, databases, software and equipment used for collecting, storing, analysing and delivering the information for managing the HR” [27]. HRIS is a database system and an efficient catalyst for linking ICT and HR management and providing the essential support for HR management [28].

HRIS facilitates and improves HR functions, policies and strategies [27]. It helps managers in HR planning, doing administrative affairs, monitoring HR activities, preparing employment and training programs and decision making [29].

C. Electronic Human Resources management (e-HRM)

E-HRM is an expression including all types of interaction between ICT and HR management [27]. His role is to create value for employees and the company [30]. E-HRM helps to align HR management with strategic objectives by integrating flexible technology. It is the delivering of HR activities enabled by information system that links candidates, workforce, managers, and the decision making. It is a means to combine HR activities and IT within company for improving human capital profitability and the enterprise performance [30].

E-HRM has many benefits such as strategic fit of HR System with HR strategy, reducing costs, saving time and budget, easy and secure access to HR information, improving efficiency and communication, improving HR services, amelioration of intellectual capital, and increasing competences and employees’ knowledge [8]. The use of e-HRM in HR functions appears in the following applications:

- **E-Recruiting and e-Selection** : The use of the Internet and other technology such as online video conferencing, to attract, select and recruit the potential candidates [8]. It refers to all kinds of technology used for assessing the candidate competencies and abilities, and attracting the most talented of them [31].
- **E-Training and e-Learning** : It covers different web-based programmes and applications used for employees learning and training such as MOOC and SPOC.
- **E-Compensation and e-Payment** : All systems used by company for tracking and managing workforce participation in benefits programmes. This process needs different information to be collected and analyzed, especially the staff absences and retards, overtime work, productivity rate and employee’s classification [8].

D. Industry 4.0 (Smart HR 4.0)

According to Hermann [32], the term “Industrie 4.0” is used for the latest industrial revolution. It became publicly known in 2011, “Industrie 4.0” allowed the communication between employees, resources and machines. Smart products know their current production, history production and target

situation. The fourth industrial revolution is characterized by a paradigm shift from centrally controlled to decentralized production processes.

Cloud Computing is a various collection of equipments and programs based on the internet and offered in real time at the request of users [33]. It is a model to allow network access to a shared computing resources (e.g. servers, applications, storage, and services) [34]. The use of industry 4.0 in HR management is called "Smart HR 4.0" [35] or "HRM 4.0" [36]. It is a new concept characterized by innovation in digital technologies such as artificial intelligence, augmented reality, cloud computing, Internet of-things [37].

III. RESEARCH METHODOLOGY

This study aims to describe the main roadmap steps of HR digital transformation. To achieve this goal, we adopted the roadmapping method.

The roadmapping is a popular method and an effective approach to strategic planning, especially for the future of technology [24]. It’s an efficacious method for studying innovation and development pathways, and provides an organized approach for mapping the evolution of emerging technologies [38]. According to Garcia et al. [39], roadmapping technology has many benefits, such as providing information about technology investment, making adequate decisions, identifying technology gaps and defining how to profit of research and development investments.

Our article aims to identify and study the main roadmaps for HR digitization. For this reason, our principal research question is the following: **“What are the essential steps of a roadmap for a successful HR digital transformation?”**

In response to this central question, we have selected seven roadmaps for identifying the main steps for a successful transformation of HR digitization. Therefore, we adopt an interpretative and constructivist approach for understanding how researchers have constructed their own roadmaps and to identify the common elements between them. In the interpretative and constructivist paradigms, the reality is never independent. The social reality is constructed by the intentions and interactions between researchers through the confrontation and the sharing of their representations [40].

IV. RESULTS AND DISCUSION

The selected HR digital transformation roadmaps can be divided in two types, specific and general roadmaps.

A. Specific roadmaps for HR digital transformation

The specific roadmaps are depending on the used technology such as cloud computing, industry 4.0 and HRIS.

1) *HR Cloud Computing implementation*: Ziebell et al. [19] made some interviews and questionnaires for collecting the necessary information from the HR professionals, he proposed seven steps for the Cloud Transformation success.

- **Digitisation strategy**

Ziebell et al. [19] mentioned some strategic reasons for HR digitization transformation: Standardization and harmonization, improving efficiency and increasing information quality.

- **The driver of digitization**

In this category he specified the individual and the department, which has driven the HR digitization process. The most cited units were the HR department, the HR director, the board of the company or a member at board level and in a few cases, the pilot of HR digital transformation was the IT department.

- **Process prioritisation**

In this part, he mentioned three categories used to prioritize the HR processes which will be digitized: Dependencies of the processes, improving on the efficiency and the latest category were hierarchy promoter. Also, he clarified that the complicated processes and those with an elevated impact are prioritized to be digitized [19].

- **Implementation Obstacles**

The main obstacles during the implementation processes were : Cultural change, resources and skills and finally technical limitations. Other obstacles were less cited such as technology costs and structure and governance.

- **HR process cluster**

This category is divided on five cluster cloud. There was a distinction between the individual HR processes, the processes digitized and the ones were in the digitizing planning. In each cluster cloud, some processes were the focus of attention (e.g. Performance and objective management was the most important process in the talent management cluster).

- **Selection criteria and implementation period**

There are several HR Cloud solutions must be looked at in advance before implementation, The most relevant selection criteria were the easy functionality of the HR cloud, and the coverage of the company needs. Other, less important criteria were economic reasons and the suppleness of changing to other technology solutions.

- **Change and evaluation**

The categories of change in HR department working methods were : The HR service efficiency and quality, and the way of HR management due to the constraint elimination and the possibilities offered by the HR Cloud. The release update is an opportunity for improving the quality of HR cloud functionality and eliminating the dysfunctions.

2) *HRMS roadmap*: Arthur [20] specify five steps for Implementing an HRMS:

- **Preparation.**

In this section, it is important to find an answer to the following strategic questions: Where will the company be in next three years? What will be the size of the company in this period? How many employees we will need? What are the internal IT skills and the need for external consultants? What is the allocated budget for the system? Where the system will be hosted (internally or externally)? Which programs will be supported (e.g. The HRIS included the

applicant tracking or not? Compensation? Employees assessment? ...).

- **Investigation**

This phase involves comparing supplier offers. It is important to prepare a requirements checklist for measuring each offer.

- **Evaluation**

In this phase, two or three providers will be identified and the decision will be prepared to make.

- **Selection**

This step will be reserved for notifying the selected provider. It is important to negotiate an additional package before signing the contract (e.g. employees training, guarantee of use and operations manual).

- **Implementation**

In the implementation step it is recommended to start out plain: Evaluated the information provided by the system, specified and customized what it cannot provide.

3) *E-HRM roadmap* : Field [21] proposed a roadmap for e-HRM implementation at the London Stock Exchange. It is composed of five steps:

- **Scope.** Linking HR and company strategy with e-HRM strategy

The HR management developed his own strategy to achieve the company's goals. The role of e-HRM was important on two distinct levels. It was a deliverable of the HR strategy and It was a mechanism for providing an HR program of change.

- **Technology.** The technology used to deliver each of the elements ; (delivering the e-HR elements)

This part, determines the technology that will support e-HRM elements, and the factors that affect the technical solution of the e-HRM deliverables. Field [21] mentioned four options for the architecture of the target system. The first option was an HR Core system with integrated modules (a single HRIS). The second was the HR Core system connected back to the existants internally modules hosted separately. The third option was the HR Core system connected back (over the web) to the existing modules hosted and managed externally. The latest option was the outsource of all systems..

- **Sequence and timing.** When the different e-HRM elements of the roadmap will be delivered; (planning the elements of the e-HRM program).

This phase is crucial to a successful integration. Field [21] defined four stages of e-HRM roadmap (Foundation, basic service, advanced service and integration). The phasing of e-HRM elements (HR activities) depends on many factors such as HR strategy, dependency to the other HR activity outcomes, complexity of the activity and the implementation cost.

- **Architecting the overall solution.** How the elements are pieced together to work as a single entity ; (developing the e-HRM roadmap)

In this part, Field [21] presented the final roadmap of e-HRM. It showed the timing of each module and has taken full regard for the different factors that may affect the quality and cost of the e-HRM element implementation.

- **Implementation plans.** Understanding the benefits, change issues and project plans.

In this part, Field [21] mentioned three types of e-HRM benefits, operational (such as reduce the global costs of HR management), relational (create a convenient access to on-line HR services) and strategic/transformational (moving HR management from an administrative function to a strategic partner).

4) *Smart RH 4.0 roadmap*: Pessl et al. [22] presented a roadmap for industry 4.0 implementation in the main company activities including HR Management.

- **Phase : Analysis**

Step 1 : Preparation (Start workshop)

In this step, it is important to create an awareness and involve HR for industry 4.0, by clarifying the fundamental components, concepts, advantages and simulation of practical cases of industry 4.0.

Step 2 : Analyse industry 4.0 - Maturity

In this phase, the company identifies the existing competencies concerning industry 4.0 and its technologic maturity level (developed capability for the principal fields of action, strategic considerations and the technology used in the operational processes).

- **Phase : Targets**

Step 3 : Define and evaluate measures

In this phase the company defines and evaluate the target state of the industry 4.0 in the action fields (organizing a workshop between the multidisciplinary expert teams for discussing the specific priorities, the existing and the future maturity level).

Step 4 : Define de target state

This phase is important to evaluate the deployed efforts, the realised benefits, the amelioration of the maturity level and the difference from the initial state. The change must be tracked and documented.

- **Phase : Realisation**

Step 5 : Prepare decision

The information about the targets maturity level and measures of industry 4.0 contribution to the company's strategy will be collected, and added to other pertinent information (e.g. key figures and real quantifiable objective)..

Step 6 : Define projects

In this step the company defines the pilot project implementation and prepare their budgets, identification and remediation of the existing

problems will be continued in this step, this operation is necessary for a good planning of the next projects.

B. General roadmaps of HR digital transformation

1) *HR digitalization roadmap proposed by Dorkenoo et al. [41]*: In their book about digital transformation, they described a roadmap for HR digitalisation composed of four steps:

- **Map skills and measure the digital maturity of the service**

Before starting the digitalization process, it is necessary to make an overview of the company: What competences and resources do you have? To define what exists, in term of digital equipment and the employees' skills, will allow to plan the first tracks and to establish the HR digitalization roadmap. It is important to organize collective reflections, in the form of brainstorming in order to identify the best digital practices, and analyse the level of the staff maturity.

- **Establish a digital transformation plan of HR management**

Define the actions to be implemented: It is the digital strategy validated by the executive committee that sets the process and achievements deadlines. This strategy can take the form of an operating manual or a table that answers the questions (what to do; who will do it; when and how to do it).

- **Train and recruit the future HR systems users**

In order to establish the digitalization operation on time and for total success, it is necessary to involve all employees and obtain their membership, and to train human resources on the methods of use and missing know-how. In the case of staff shortage, it is important to start the recruitment process for selecting the desired digital skills.

- **Digital transformation assessment and consolidation.**

It is important to assess digital transformation actions in order to know whether or not the objectives initially planned were achieved. If the objectives are not fully achieved, it is indispensable to plan the areas for improvement for optimizing the processes.

2) *HR digitalisation roadmap proposed by Storhaye [23]*: He established an HR digital roadmap composed of four themes.

- **HR process & culture**

It is important to integrate digital culture and its tools into HR management practices, including those that seem the most complex or the least favorable. HR practitioners must have the necessary culture and skills in order to know if the company is strongly committed to this transformation.

- **Work & organization**

The managers must ensure that the company can benefit from the transformations that digital technology allows both in terms of working way and of organizational and managerial culture. It is

necessary to know what the technologies make possible as new working methods and all the consequences on HR management and effects on the quality of life at work.

- **Relations with the ecosystem**

The company must have the capacity to adapt to its environment and the changes it manifests as digital becomes a norm. It is necessary to adapt HR tools and processes to the requirements of external actors.

- **Business, strategy and HR**

Company managers must help strengthen the role that digital must play in the business model and strategy of the company. They must supervise the HR digitalization consequences on the jobs evolution, skills and culture.

The adoption of digital revolutionizes the way of exercising a profession, so professional behaviour must also change because expectations change.

3) *HR digitalisation roadmap proposed by Chaintreuil [42]*: He offered a plan for HR digitalisation has four points:

- **Acculturation: How to best position the basis of a digital culture?**

The executive committee must lay the foundations for common learning and a new digital culture, which will allow to change the way of thinking, to innovate and a better management of the digital transformation.

- **A relevant digital culture for a successful digital transformation**

Digital transformation is based on the appropriation and dissemination of a real digital culture in the company. Also, it's based on collaborative intelligence, openness to an ecosystem, open innovation, constant progress in working methods and trades and the new participation of talents

- **From a corporate culture to a digital corporate culture**

The introduction of digital culture in the company requires reassuring employees by showing them that it will not completely replace the company's culture but will be incorporated into it. The managers must also explain to them what they will gain from the digital transformation.

- **The importance of digital skills**

The training makes it possible to involve employees at all levels in digital transformation, so that they integrate digital skills. These include digital literacy, digital project management and digital leadership.

TABLE I. COMMON ELEMENTS IN DIGITAL HR ROADMAPS

Common features	Study reference
Digital Skills	[19] ; [20] ; [22] ; [23] ; [41] ; [42]
Digital strategy	[20] ; [21] ; [22] ; [23] ; [41]
Adopted technology	[19] ; [20] ; [21] ; [22]
Allocated budget and costs	[19] ; [20] ; [21] ; [22]
Measures evaluation	[19] ; [20] ; [22] ; [41]
Company culture	[19] ; [23] ; [42]
Digital maturity	[22] ; [41]

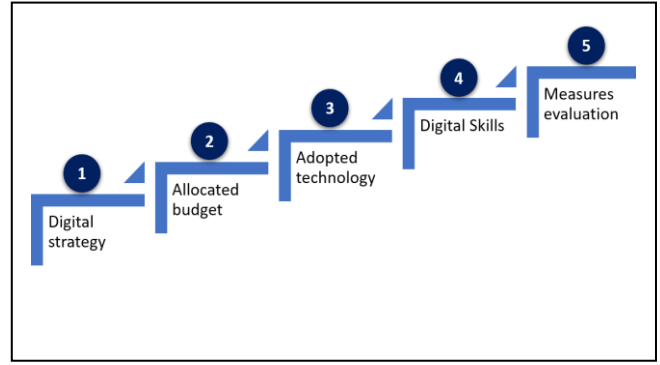


Fig. 1. The proposed roadmap for the digital transformation of HR

C. The main common steps between the selected roadmaps

- **Digital Skills**

The success of an HR digital transformation roadmap depends principally on the active participation of human capital who has adequate digital skills [19]. It includes communication and sharing, information management, technical operations, creation of content and knowledge, collaboration and problem solving [43].

- **Digital Strategy**

The digital HR strategy is the fusion of HR and ICT for generating the best advantages and the great challenges for the organization. It is an HR strategy developed and realized by using digital technology to create value [7].

- **Adopted technology**

The choice of the appropriate technology depends on several factors (The size of the company, the cost of the program and the employee skills). The adopted technology can be used in several architectures (A single HRIS, HR system with modules implemented separately on internal networks, HR system with modules hosted externally) [21].

The **Table I** presents the main common steps mentioned in the analyzed roadmaps.

The common steps between the selected roadmaps allowed us to establish our own roadmap, as shown in **Figure 1**.

The limitations of this study result from the number of the selected roadmaps, it does not exceed seven roadmaps (four are specific to some HR systems, the others are general). The obtained results cannot be generalized to all types of organizations.

V. CONCLUSION

The proposed paper clarifies that the Covid-19 pandemic has accelerated the use of ICT and the digital transformation of organizations. Therefore, those that are digitally advanced have been able to continue their activities normally, and among these the HR management.

This paper identifies the main HR Digitalization types (e-HRM, Smart HR 4.0, HRIS, ...), its contribution in HR management, and provide a general overview of roadmap concept.

The document allows to analyze some HR digital transformation roadmaps and aims to determine the principal steps and the essential elements of a successful implementation of a roadmap in this filed.

Finally, the paper presents a roadmap for the digital transformation of HR that is constituted of the common steps proposed by researchers.

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